

**This assignment will not be graded and is only for practice.**

### Key points: 1

The span of a set  $S$  (of vectors) is defined as the set of all finite linear combinations of elements (vectors) of  $S$  and denoted by  $\text{Span}(S)$  i.e.,  $\text{Span}(S) = \{\sum_{i=1}^n a_i v_i \in V \mid a_1, a_2, \dots, a_n \in \mathbb{R} \text{ and } v_1, v_2, \dots, v_n \in S\}$ .

Let  $V$  be a vector space. A set  $S \subseteq V$  is spanning set for  $V$  if  $\text{Span}(S) = V$ .

### Algorithm to obtain a spanning set for a non-zero vector space $V$ :

- i) Start with the empty set  $S_0 = \phi$ .
- ii) Add any non-zero vector  $v_1 \in V$  to  $S_0$  and define it as  $S_1$ . So  $S_1 = S_0 \cup \{v_1\}$ .
- iii) If  $\text{Span}(S_1) = V$ , then we are done. Otherwise choose  $v_2 \in V \setminus \text{span}(S_1)$  and add it to the set and call it  $S_2$ . So  $S_2 = S_1 \cup \{v_2\}$ .
- iv) If  $\text{Span}(S_2) = V$ , then we are done. Otherwise, choose  $v_3 \in V \setminus \text{Span}(S_2)$ , and construct  $S_3 = S_2 \cup \{v_3\}$ .
- v) Repeat the process until  $\text{Span}(S_n) = V$  for some  $n$ .

1) Let  $S = \{(1, 2), (2, 3), (1, -1)\} \subseteq \mathbb{R}^2$ . Which of the following is (are) elements of  $\text{Span}(S)$ ? **1 point**

- ☐  $5(1,2) + 8(1,-1)$
- ☐  $(0,0)$
- ☐  $2(1, 2) - 3(2, 3) + 5(1, -1)$
- ☐  $(2, 1) - 2(3, 2) - 4(-1, 1)$

No, the answer is incorrect.

Score: 0

### Accepted Answers:

$5(1,2) + 8(1,-1)$   
 $(0,0)$   
 $2(1, 2) - 3(2, 3) + 5(1, -1)$   
 $(2, 1) - 2(3, 2) - 4(-1, 1)$

2) Let  $S = \{(1, 1, -1), (-1, 1, 1)\} \subset \mathbb{R}^3$ . Which of the following is (are) elements of  $\text{Span}(S)$ ? **1 point**

- ☐  $2(1, 1, -1) + (-1, 1, 1)$
- ☐  $(8, 8, -8)$
- ☐  $(0, 2, 0)$

☐ (1,1,-2)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$2(1, 1, -1) + (-1, 1, 1)$

$(8, 8, -8)$

$(0, 2, 0)$

3) Let  $S = \{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\} \subset \mathbb{R}^3$ . Which of the following option(s) is (are) true? **1 point**

[Hint: Form the matrix using the vectors as the columns and calculate the determinant of the matrix.]

- ☐  $S$  is a linearly dependent set.
- ☐  $S$  is a linearly independent set.
- ☐  $S$  is a basis of the vector space  $\mathbb{R}^3$ .
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S$  is a linearly independent set.

$S$  is a basis of the vector space  $\mathbb{R}^3$ .

4) Consider a set of vectors  $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$  in  $\mathbb{R}^2$ . Choose the set of correct options. **1 point**

- ☐ The set  $\{(1, 2), (-1, 2), (3, 1)\}$  is a basis of  $\mathbb{R}^2$ .
- ☐ The set  $\{(-1, 2), (3, 1)\}$  is a basis of  $\mathbb{R}^2$ .
- ☐ The set  $\{(3, 1), (-3, 1)\}$  is a basis of  $\mathbb{R}^2$ .
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

**Accepted Answers:**

The set  $\{(-1, 2), (3, 1)\}$  is a basis of  $\mathbb{R}^2$ .

The set  $\{(3, 1), (-3, 1)\}$  is a basis of  $\mathbb{R}^2$ .

**Key points: 2**

The following conditions are equivalent for a set to be a basis of a vector space:

- Maximal linearly independent set : A linearly independent set such that if any vector is appended to it, the new set is no longer linearly independent.

- Minimal spanning set : A spanning set such that if any vector is removed from the set, then the set will not be a spanning set.

5) Which of the following sets are maximal linearly independent sets in  $\mathbb{R}^2$ ?

**1 point**

- ☐  $\{(1, 0)\}$
- ☐  $\{(1, 2), (2, 4)\}$
- ☐  $\{(1, 2), (1, 1)\}$
- ☐  $\{(1, 2), (1, 1), (2, 1)\}$
- ☐  $\{(1, 0), (0, 1)\}$

No, the answer is incorrect.

Score: 0

**Accepted Answers:**

$\{(1, 2), (1, 1)\}$

$\{(1, 0), (0, 1)\}$

6) Which of the following sets are minimal spanning sets of  $\mathbb{R}^3$ ?

**1 point**

- ☐  $\{(1, 0, 0), (0, 1, 0)\}$
- ☐  $\{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$
- ☐  $\{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$

- ☐  $\{(1, 0, 1), (0, 1, 1), (1, 1, 2)\}$
- ☐  $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$   
 $\{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$

### Key Points: 3

**Basis:** A basis of a vector space  $V$  is a linearly independent subset of  $V$  that spans  $V$ .

Basis of a vector space:

- Maximal linearly independent set.
- Minimal spanning set.

How to find a basis of a subspace of  $\mathbb{R}^n$ ?

- Suppose a subspace  $V$  of  $\mathbb{R}^n$  is defined by some constraints, as follows:

$$V = \{(x_1, x_2, \dots, x_n) \mid f_1(x_1, x_2, \dots, x_n) = 0, f_2(x_1, x_2, \dots, x_n) = 0, \dots, f_m(x_1, x_2, \dots, x_n) = 0\} \subseteq \mathbb{R}^n$$

Where  $f_i$ 's are linear equations in general.

- **Step 1:** By solving  $f_i$ 's we have to try to find out the dependent and independent variables.

Suppose there are  $k$  independent variables

and there are  $k - n$  dependent variables.

- **Step 2:** Choose the first independent variable, let it be  $x_1$  and assign the value 1 to that, and assign 0 to all the other independent variables.

- **Step 3:** Find out the values of all the dependent variables with respect to the assignment given in Step 2 and call that vector to be  $v_1$ .

- **Step 4:** Carry out this process for each independent variable, to get the vectors  $v_2, \dots, v_k$ .

The set of vectors  $\{v_1, v_2, \dots, v_k\}$  is a linearly independent set and this is a maximal linearly independent set in  $V$ , hence forms a basis of  $V$ . Hence, dimension of  $V$  is  $k$  in this case.

7)

**1 point**

Let  $S = \{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\} \subset \mathbb{R}^3$ . Which of the following option(s) is (are) true?

**Hint:** Form the matrix using the vectors as the columns and calculate the determinant of the matrix.

- ☐  $S$  is a linearly dependent set.
- ☐  $S$  is a linearly independent set.
- ☐  $S$  is a basis of the vector space  $\mathbb{R}^3$ .
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S$  is a linearly independent set.

$S$  is a basis of the vector space  $\mathbb{R}^3$ .

8) Consider a set of vectors  $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$  in  $\mathbb{R}^2$ . Choose the set of correct options.

**1 point**

- ☐ The set  $\{(1, 2), (-1, 2), (3, 1)\}$  is a basis of  $\mathbb{R}^2$ .
- ☐ The set  $\{(-1, 2), (3, 1)\}$  is a basis of  $\mathbb{R}^2$ .
- ☐ The set  $\{(3, 1), (-3, 1)\}$  is a basis of  $\mathbb{R}^2$ .
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set  $\{(-1, 2), (3, 1)\}$  is a basis of  $\mathbb{R}^2$ .

The set  $\{(3, 1), (-3, 1)\}$  is a basis of  $\mathbb{R}^2$ .

9) Let  $V$  be a vector space which is defined as follows:  $V = \{(x, y, z) \mid 2x + 3y = 0 \text{ and } y + 5z = 0\} \subseteq \mathbb{R}^3$  with usual addition and scalar multiplication. Which of the following sets form bases of  $V$ ? (More than one option may be correct) **1 point**

[Hint: Two constraints are given here,  $f_1(x, y, z) = 2x + 3y = 0$  and  $f_2(x, y, z) = y + 5z = 0$ . Observe that  $z$  is the independent variable and  $x$  and  $y$  are the dependent variables.]

- ☐  $\{(2, 3, 0), (0, 1, 5)\}$
- ☐  $\{(3, -2, 0), (0, 5, -1)\}$
- ☐  $\{(\frac{15}{2}, -5, 1)\}$
- ☐  $\{(15, -10, 2)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(\frac{15}{2}, -5, 1)\}$   
 $\{(15, -10, 2)\}$

10) Let  $V$  be a vector space which is defined as follows:  $V = \{(x, y, z, w) \mid x + z = y + w\} \subseteq \mathbb{R}^4$  with usual addition and scalar multiplication. Which of the following set forms a basis of  $V$ ? **1 point**

[Hint: The constraint given here is  $x + z = y + w$ , which can be written as  $x - y + z - w = 0$ . As there are 4 variables and 1 equation, the number of independent variables is 3, where as the number of dependent variable is 1. If you check the first vector of each option, only Option 2 and Option 4 satisfying the constraint, and in that vector  $x = y = 1$ . Start with taking  $y$  as the dependent variable, and apply the method mentioned above.]

- ☐  $\{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 1)\}$ .
- ☐  $\{(1, 1, 0, 0), (0, 1, -1, 0), (0, -1, 0, 1)\}$ .
- ☐  $\{(1, -1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$ .
- ☐  $\{(1, 1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$ .

11) Find the dimension of the vector space  $V = \{(x, y, z, w) \mid x + y = z + w, x + w = y + z, \text{ and } x, y, z, w \in \mathbb{R}\}$ . [Hint: Two constraints are given here  $f_1(x, y, z, w) = x + y - z - w = 0$  and  $f_2(x, y, z, w) = x - y - z + w = 0$ . Try to find out the number of independent variables.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

### Key Points: 4

How to find a basis of a subspace  $V$  of  $M_{m \times n}(\mathbb{R})$ :

- **Step 1:** Consider each element of a  $m \times n$  matrix as a variable. So there are  $mn$  number of variables.
- **Step 2:** Using the constraints given as the definition of the subspace separate out the independent and the dependent variables. Suppose there are  $k$  independent variables and  $mn - k$  dependent variables.
- **Step 3:** Define a matrix  $M_1$  by assigning 1 to one independent variable and 0 to all the other independent variables, together with finding out the values of all the dependent variables.
- **Step 4:** Repeat this process for all the independent variables to get the vectors  $M_2, M_3, \dots, M_k$ .

The set of matrices  $\{M_1, M_2, \dots, M_k\}$  forms a basis of  $V$ . Hence the dimension of  $V$  is  $k$  in this case.

12) Let  $V = \{A \mid A \in M_{3 \times 3}(\mathbb{R}), A \text{ is a diagonal matrix}\}$ . Which of the following sets form a **1 point** basis of  $V$ ?

[Hint:  $a_{ij} = 0$  if  $i \neq j$ . Thus there are 6 constraints and these are the dependent variables while the diagonal entries are the 3 independent elements.]

- ☐  $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$
- ☐  $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$
- ☐  $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$
- ☐  $\left\{ \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \right\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

13) Let  $V = \{A \mid A \in M_{3 \times 3}(\mathbb{R}), A \text{ is a diagonal matrix and sum of the diagonal elements are } 0\}$ . What will be the dimension of  $V$ ?

[Hint: The general form a diagonal matrix of order 3 is  $\begin{bmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{bmatrix}$ . Hence there are three variables,

$a_1, a_2, a_3$ . Now there is one more constraint, which says that the sum of the diagonal elements is 0. Hence we have  $a_1 + a_2 + a_3 = 0$ . So we can choose  $a_3$  to be the dependent variable and  $a_1$  and  $a_2$  to be the independent variables, as  $a_3 = -a_1 - a_2$ .]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

### Key points: 5

Rank of a matrix:

- Maximum number of linearly independent column vectors (or row vectors) in a matrix.

NOTE: Rank of a matrix is always less than or equal to the number of columns (or rows).

How to find rank of a matrix A:

- Reduce the matrix to row echelon form.
- Find the number of non zero rows in the reduced matrix which will be the rank of the matrix A.

14) Consider the following two matrices:

**1 point**

$$P = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -2 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 1 \end{bmatrix}; Q = \begin{bmatrix} 1 & 1 & -2 & 0 \\ 1 & 0 & 2 & 0 \\ 0 & -1 & 0 & 4 \end{bmatrix}$$

Which of the following statements is (are) TRUE? [ Hint: Try to reduce the matrices to row echelon form and find the number of non zero rows ]

- ☐ Rank(P) = 4
- ☐ Rank(P) = Rank(Q)
- ☐ Rank(QP) = 4
- ☐ Rank(PQ) = Rank(QP)

No, the answer is incorrect.

Score: 0

Accepted Answers:

Rank(P) = Rank(Q)

$$\text{Rank}(PQ) = \text{Rank}(QP)$$

**Your score is: 0/14**